

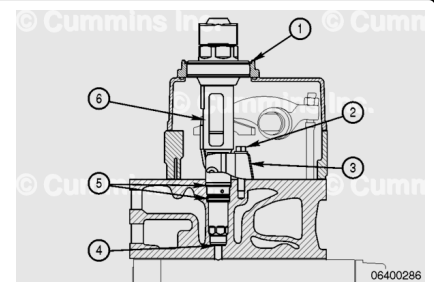
Trainings ▾

General Information

with Electronically Actuated Injector

The injector consists of the following components:

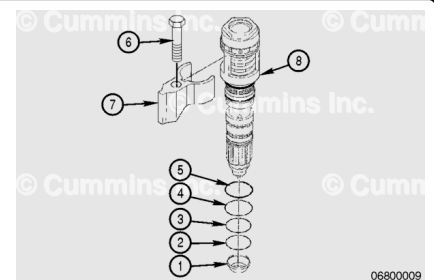
1. O-ring and damper
2. Capscrew
3. Injector hold-down clamp
4. Seal
5. O-rings
6. Injector.



with Mechanically Actuated Injector

The injector consists of the following components:

1. Sealing ring
2. O-ring (black, small)
3. O-ring (green)
4. O-ring (blue)
5. O-ring (black, large)
6. Capscrew
7. Injector hold down clamp
8. Injector.



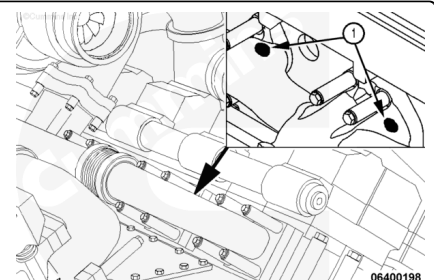
Initial Check

Misfire Using Heat Indicator Marker

Note : This check applies only to mechanically actuated injectors.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam



can cause personal injury.

Note : This procedure is **not** effective when the symptoms occur **only** at low idle or when the engine is cold.

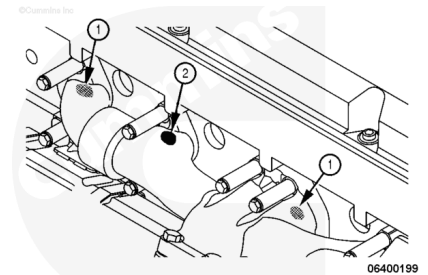
If the engine exhibits symptoms of an injector **not** firing, perform the following check:

- Operate the engine to normal operating coolant temperature of 80°C [180°F].
- Stop the engine and allow the exhaust manifold to cool for 10 minutes.
- Use a 204°C [400°F] temperature indicator marker, Part Number 3165163, or equivalent, to apply a mark (1) on the surface of all exhaust manifold ports. The mark **must** be as close to the cylinder head as possible, but **not** directly on the exhaust manifold mounting flange.
- Start the engine and operate under light load for a short time. Do **not** operate the engine under heavy load or for an extended period of time, as this can cause inaccurate results.

Note : It is **not** recommended to use the temperature indicator markers to troubleshoot for cylinders that are operating too hot.

If the 204°C [400°F] mark melts (1) the color will disappear and leave **only** a transparent glazed appearance where the mark was. This indicates that the cylinder is operating at a temperature above 204°C [400°F].

If the 204°C [400°F] mark does **not** melt (2) the mark color remains present. This indicates that the cylinder is **not** operating at a temperature of 204°C [400°F], indicating a possible injector malfunction.

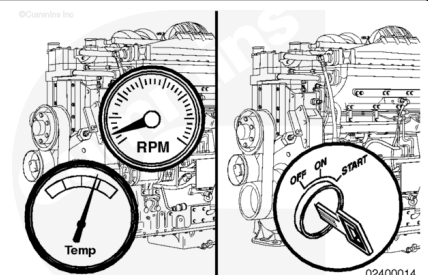


Misfire Using Injector Cutout Tool

Note : This test applies **only** to mechanically actuated injectors.

⚠ CAUTION ⚠

This tool is not intended to be used above the engine Idle speed. Do not use above 900 rpm engine speed. Doing so can damage the engine. Start and operate the engine until

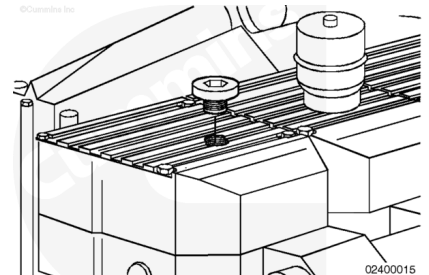


the coolant temperature reaches the normal range of 80°C [180°F]. Shut the engine off.

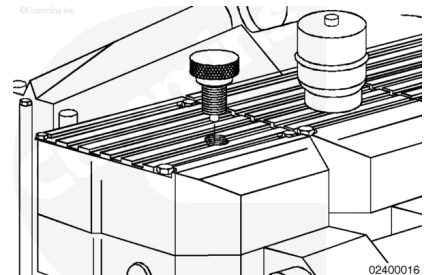
Start and operate the engine until the coolant temperature reaches the normal range of 80°C [180°F].

Shut the engine off.

Use a 12-mm hex socket or allen wrench to remove the existing plug from the top side of the cast iron valve cover.



Screw the single injector diagnostic cut-out tool **clockwise** into the threaded hole in the valve cover, until it bottoms out on the valve cover.



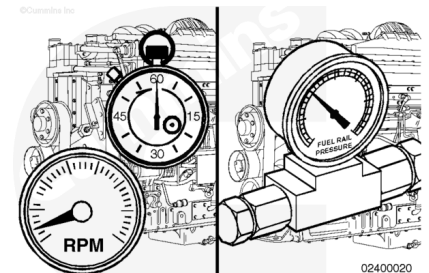
⚠ CAUTION ⚠

This tool is not intended to be used above the engine idle speed. Do not use above 900 rpm engine speed. Doing so can damage the engine.

Install a 0 to 207 kPa [0 to 30 psi] pressure gauge on the Compuchek™ fitting located at the front of the block.

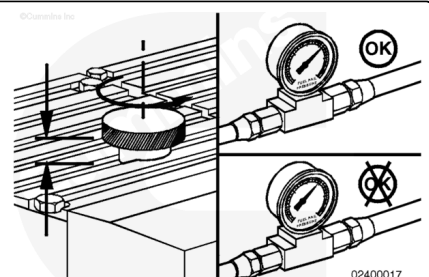
Start and operate engine at idle speed for approximately 1 minute.

1. Record the fuel rail pressure reading.



Rotate the single-injector diagnostic cut-out tool **counterclockwise** until it is approximately 13 mm [½ inch] clear of the valve cover.

2. Note the fuel rail pressure reading.



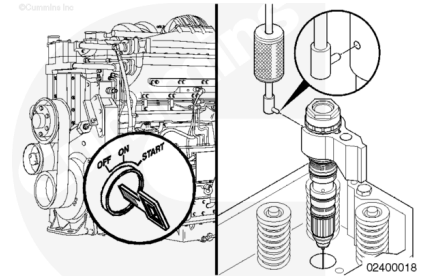
If the cylinder being tested is working properly, the fuel rail pressure can decrease on reading number 2.

If the cylinder being tested is weak or **not** firing, the fuel rail pressure will **not** change between reading number 1 and number 2.

Shut off the engine.

Repeat the above steps in different rocker lever covers until the faulty injector is found.

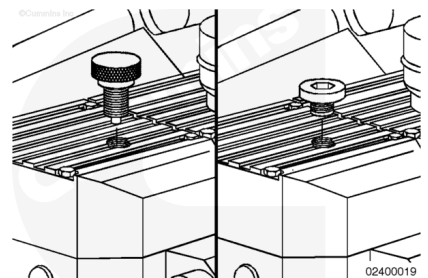
Replace the faulty injector(s).



After engine diagnostic testing, remove the single-injector diagnostic cut-out tool from the valve cover and install the plug.

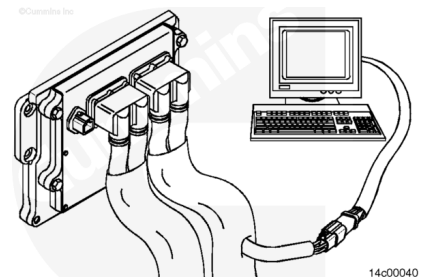
Tighten the plug.

Torque Value: 45 n·m [33 ft-lb]



Electronically Actuated Injector

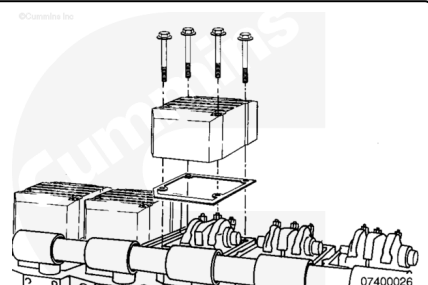
Use INSITE™ electronic service tool to perform the cylinder cutout test on engines with electronically actuated injectors. Reference the INSITE™ electronic service tool service manual for instructions.



Check the injectors (mechanically actuated) when it is suspected they were run dry (without fuel).

Remove the rocker lever covers and check to be sure all of the injector links are intact. If they are intact, install the covers and follow the normal injector inspection procedure. If they are **not** intact, follow the injector inspection step in this procedure.

Warm up the engine by running at high idle to stabilize the exhaust temperature. If the exhaust temperatures are within 38°C [100°F], all injectors are firing and working correctly.



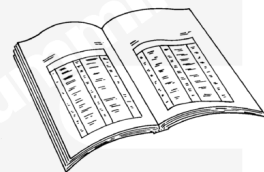
Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

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ck800wa

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

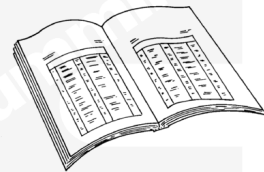
- Remove the rocker lever cover. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-011.html)
- Remove the rocker lever assembly. Refer to Procedure 003-009 (Rocker Lever Assembly) in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-009.html)

with Electronically Actuated Injector

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ck800wa

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

- Remove the injector supply lines. Refer to Procedure 006-051 (Injector Supply Lines (High Pressure)) in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Install the injector protective plugs, Part Number 4918767.
- Remove the two-piece valve cover. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3. (</qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html>)

Remove

with Mechanically Actuated Injector

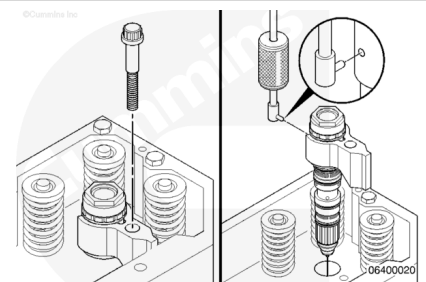
Note : The hold-down clamp can **not** be removed until the injector is removed.

Remove the injector hold-down capscrew. Remove the injector and clamp.

Use injector puller, Part Number 3823579 or equivalent, to remove the injectors. Insert the pin of the tool in the hole provided in the body of the injector. The hole faces the front of the engine.

Note : If the injector puller is **not** available, carefully use a pry bar. Pry upward on the injector against the cylinder head.

Cover the injection bore to prevent foreign objects from entering the cylinder.

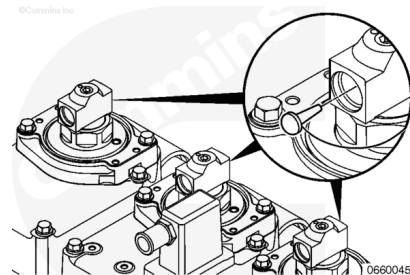


with Electronically Actuated Injector

Note : Do not reuse protective plugs. The protective plugs **must** be used immediately upon removal from the plastic wrapping. If they are **not** used immediately, the protective plugs **must** be discarded. If the protective plugs are fouled in any manner before use,

the protective plugs **must** be discarded and a new set used.

If **not** already completed, plug the ports in each side of the injector tee-piece with the injector protective plugs before removing the injector. Make sure that the plugs are wedged firmly into place. This will prevent debris from entering and causing damage to the injector.



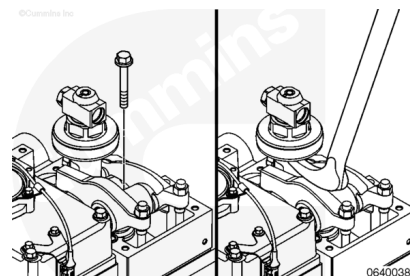
⚠ CAUTION ⚠

DO NOT reuse the Injector hold down capscrew or Injector combustion seal. The reuse of those components can result in engine damage.

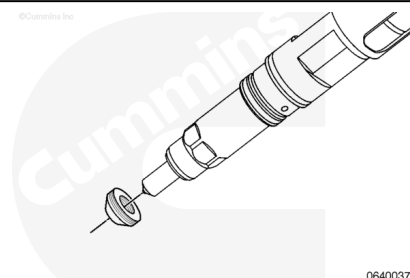
Remove and discard the injector hold-down capscrew.

Use a heel bar to remove the injector. Place the heel between the injector body and the exhaust rocker lever and gently pry the injector from the injector bore.

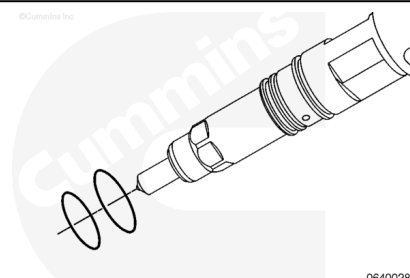
Cover the injector bore in the cylinder head to prevent foreign objects from entering and causing damage to the bore and combustion chamber.



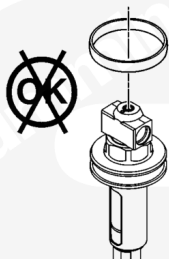
Remove and discard the seal on the end of the injector.



Remove and discard the two o-rings on the injector.

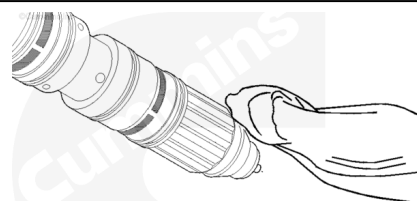


Do **not** remove the o-ring/damper at this time.



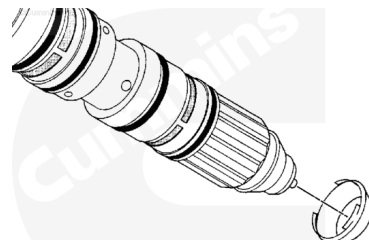
Clean and Inspect for Reuse with Mechanically Actuated Injector

Use a lint free cloth to clean the exterior of the injector.



Sealing rings are available in different thicknesses to adjust the injector protrusion.

Remove the sealing ring and note the cylinder location.



The four new o-rings **must** be installed and oriented correctly on the injector. Identify the o-rings to install them in the correct grooves.

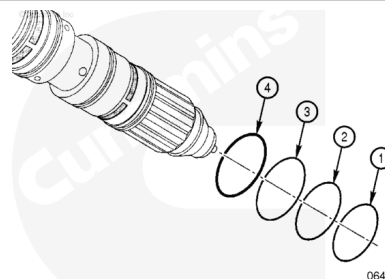
Remove the o-rings (1, 2, 3, and 4).

Use a lint-free cloth. Clean the exterior of the injector.

Carefully check the area where the sealing ring touches the injector.

Inspect the o-ring grooves for damage.

Inspect the injector body and cup retainer for cracks or other damage.

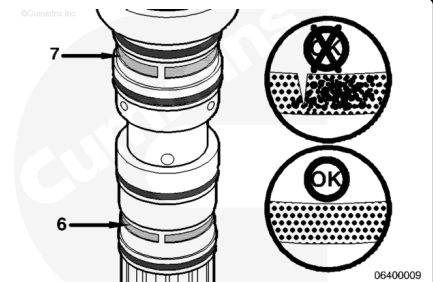


Inspect the injector links for damage, excessive wear, and pitting or scoring on the ball ends.

Note : If the link is damaged, or if pitting or scoring can be seen or felt, the link must be replaced.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

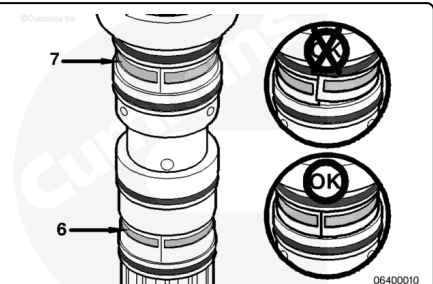


Use solvent when cleaning the filters. The screens **must** be thoroughly dry prior to installation.

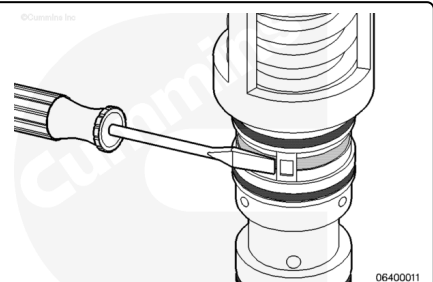
Check the rail (6) and timing (7) filter screens for debris, tears, or punctures. Replace the filter screens if damaged or dirty.

Check the rail (6) and timing (7) filter screens for proper installation.

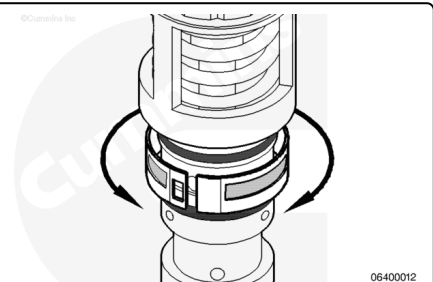
The filter screens **must** be securely snapped into the correct position.



Use a small screwdriver to remove the filter screen by gently prying up on the connector clip.

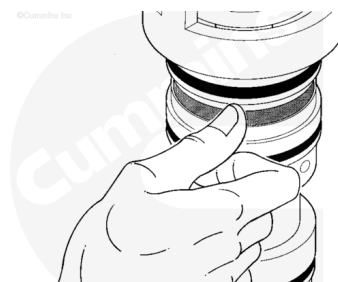


Wrap the filter screen around the inlet groove.



A properly secured connector will create a single snapping sound.

Apply pressure on the filter screen connector clip.

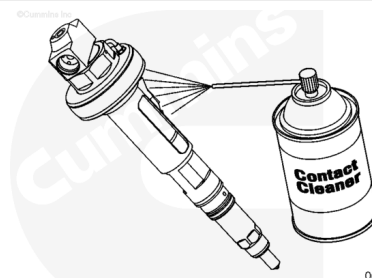


06400013

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



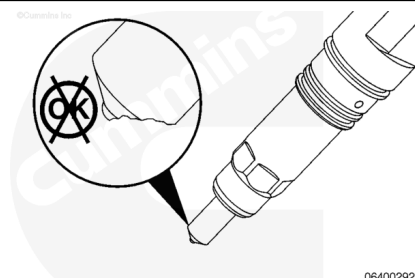
06400291

Clean the injector assembly with contact cleaner, Part Number 3824510 or equivalent. Allow to air dry completely.

Do **not** use compressed air to clean or dry the injector.

Inspect the nozzle tip for damage.

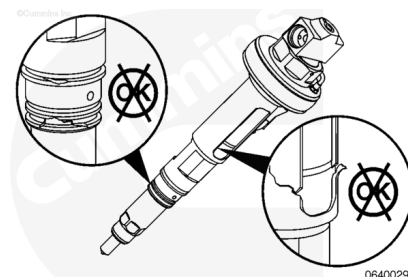
Replace injectors with damaged nozzles that prevent fuel delivery into the combustion chamber.



06400292

Inspect the injector body for signs of corrosion, cracks, or other damage.

Inspect the o-ring grooves for damage that will **not** allow the o-rings to seal properly.

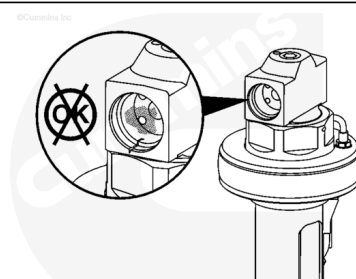


06400293

Inspect the t-fitting on top of the injector where the injector supply lines attach.

Inspect for corrosion, cracks, or other damage that will **not** allow proper sealing of the injector supply lines.

Replace damaged injectors.

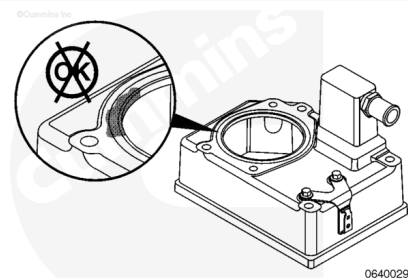


06400294

Inspect the o-ring/damper for wear, cracks, or other damage. Replace, if damaged.

If the valve cover shows signs of oil leakage around the injector, replace the o-ring/damper.

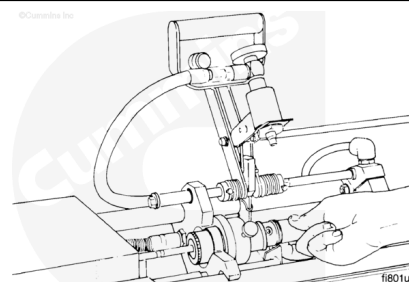
Inspect the injector wiring for damaged wires, shrink tubing, elbow, and connector.



Calibrate

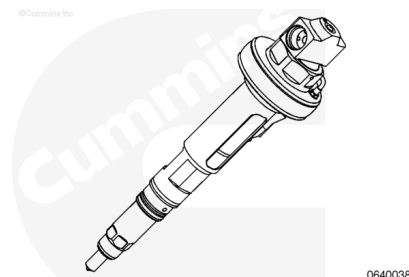
with Mechanically Actuated Injector

This procedure requires special equipment and **must** be done at a Cummins® Authorized Repair Location.



with Electronically Actuated Injector

Calibration is **not** required for electronically actuated injectors.



Install

with Mechanically Actuated Injector

Do **not** lubricate the o-rings until the injector is ready for installation in the cylinder head.

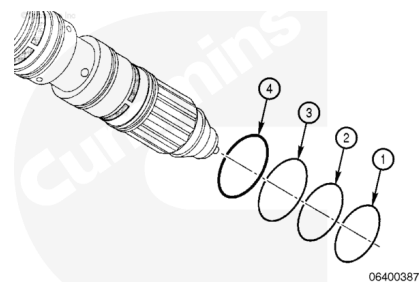
Install four new o-rings on the injector. Do **not** twist the o-rings.

O-ring (1) is brown and is smaller in diameter than the other three.

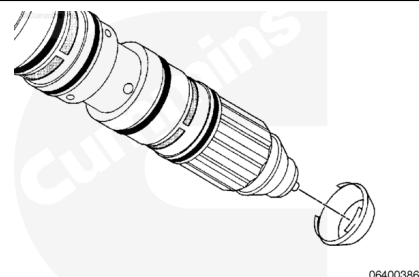
O-ring (2) is black.

O-ring (3) is brown and is larger in diameter than (2), but has a smaller cross section than o-ring (4).

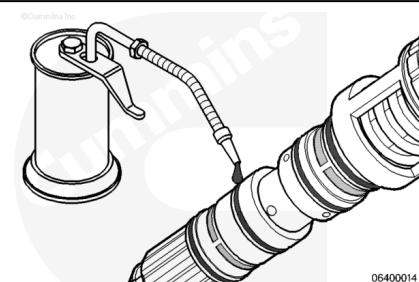
O-ring (4) is brown and is the largest in diameter and cross section.



Install the proper size sealing ring.



Use clean engine lubricating oil (15W-40) to lubricate the o-rings.



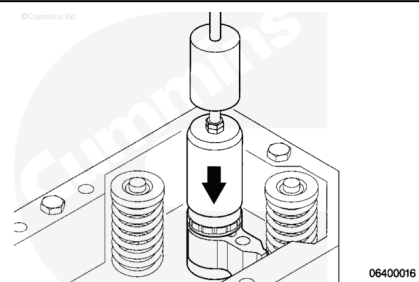
Install the hold-down clamp on the side of the injector.

Install the injector and hold-down clamp in the cylinder head.

Note : If the injector installation tool, Part Number 3824830, is not available, use a 40 mm [1 9/16 inch] socket with an extension and a rubber mallet to install the injector.

Use injector installation tool, Part Number 3824830, to set the injector in the bore.

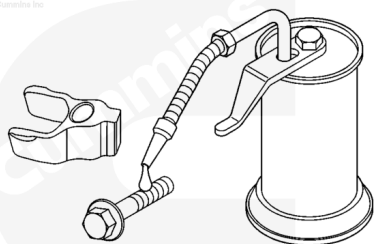
The slide hammer will make a dull sound when the injector is seated properly.



Use clean engine oil to lubricate the injector hold-down capscrew.

Install and tighten the injector hold-down capscrew.

Torque Value: 75 n•m [55 ft-lb]

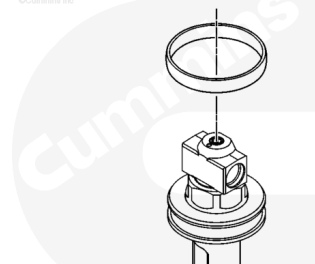


06400365

with Electronically Actuated Injector

Install the new o-ring/damper on the injector.

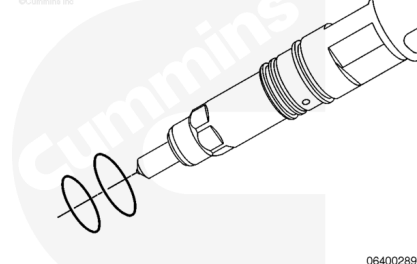
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06800270

Install the new o-rings on the injector body.

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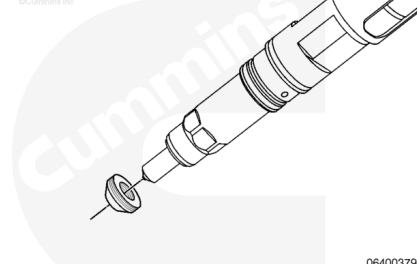


06400289

⚠ CAUTION ⚠

DO NOT reuse the Injector hold down capscrew or Injector combustion seal. The reuse of those components can result in engine damage.

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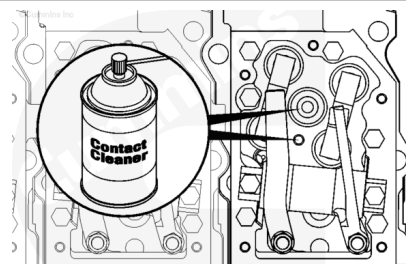


06400379

Install the new seal on the end of the injector.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.



06400364

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Keep the injector bore and combustion chamber protected when using compressed air to prevent contamination. Failure to do so can cause engine damage.

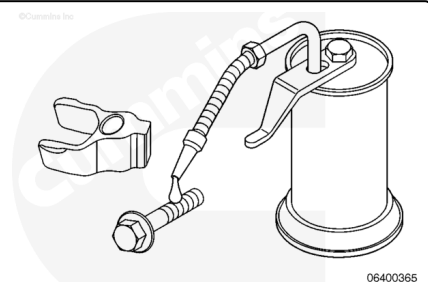
⚠ CAUTION ⚠

DO NOT reuse the Injector hold down capscrew or Injector combustion seal. The reuse of those components can result in engine damage.

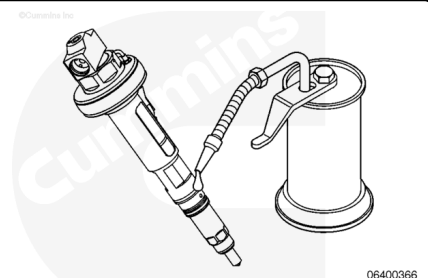
Clean the injector bore in the cylinder head with contact cleaner, Part Number 3824510 or equivalent, for proper sealing of the o-rings.

Clean the threaded hole in the cylinder head where the injector hold-down capscrew will be installed with contact cleaner, Part Number 3824510 or equivalent, and compressed air.

Lubricate the threads and underside of the new injector hold-down capscrew with clean engine lubricating oil.

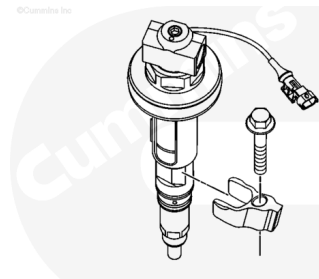


Lubricate all of the o-rings with a light coating of clean engine lubricating oil.



⚠ CAUTION ⚠

DO NOT reuse the Injector hold down capscrew or Injector combustion seal. The reuse of those components can result in engine damage.

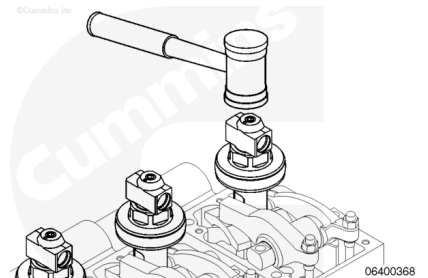


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Assemble the injector hold-down clamp and capscrew into the injector.

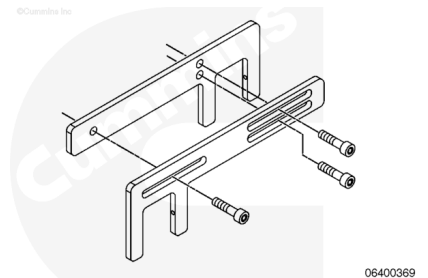
The injector installer can be used on single or multiple injector installations.

Install the assembly into the cylinder head by tapping gently on the injector with a soft face mallet.



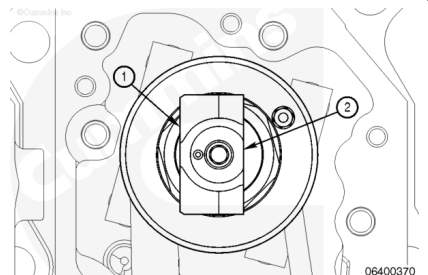
06400368

Assemble the injector installer kit, Part Number 4918620. Finger tighten the three shoulder capscrews.



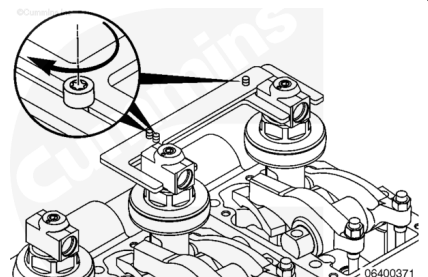
06400369

Place the injector installer on the injector such that the ball spring plunger is on the tapered side of the injector (1). If necessary, adjust the ball spring plunger so that it presses against the injector. Make sure the flat side of the injector (2) sits flush with the injector installer.



06400370

Align the injector installer with two injectors.



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Tighten the injector hold-down capscrews.

The 60 degree torque angle can be achieved at the capscrew by marking the location of one corner of the capscrew to a location on the injector hold-down clamp.

Tighten to:

Torque Value: 68 n•m [50 ft-lb]

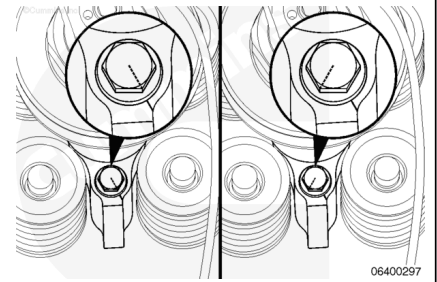
Back off completely.

Tighten to:

Torque Value: 30 n•m [22 ft-lb]

Rotate the capscrew 60 degrees.

Remove the injector installer.



Finishing Steps

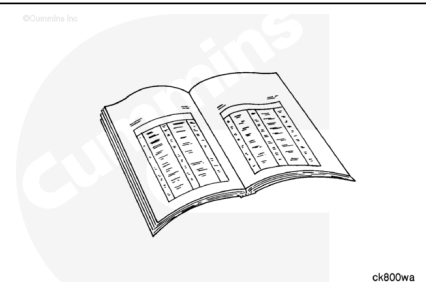
with Mechanically Actuated Injector

- Install the rocker lever assembly. Refer to Procedure 003-009 (Rocker Lever Assembly) in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-009-tr.html)
- Adjust the valves and injectors. Refer to Procedure 003-006 (Overhead Set (OBC)) in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Operate the engine and check for leaks.



with Electronically Actuated Injector

- Adjust the valves. Refer to Procedure 003-006 (Overhead Set (OBC)) in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 (Rocker Lever Cover) in Section 3.
(/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Remove the protective plugs from the injectors.



- Install the injector supply lines. Refer to Procedure 006-051 (Injector Supply Lines (High Pressure)) in Section 6. (</qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html>)
- Operate the engine and check for leaks.

Last Modified: 29-Jan-2008
